



Yield Optimization - Process Industry

Anurag Seksaria

Process Parameters Under Study

Batch Number

A1 %

A2 %

A3%

Coarse

Coarse TPH

Fine

Fine TPH

Hold Time

Fill Temp

Temp-2

Dosing Rate

A4 LPH

Objective & Challenges

Maximize Yield while maintaining the Quality and Throughput

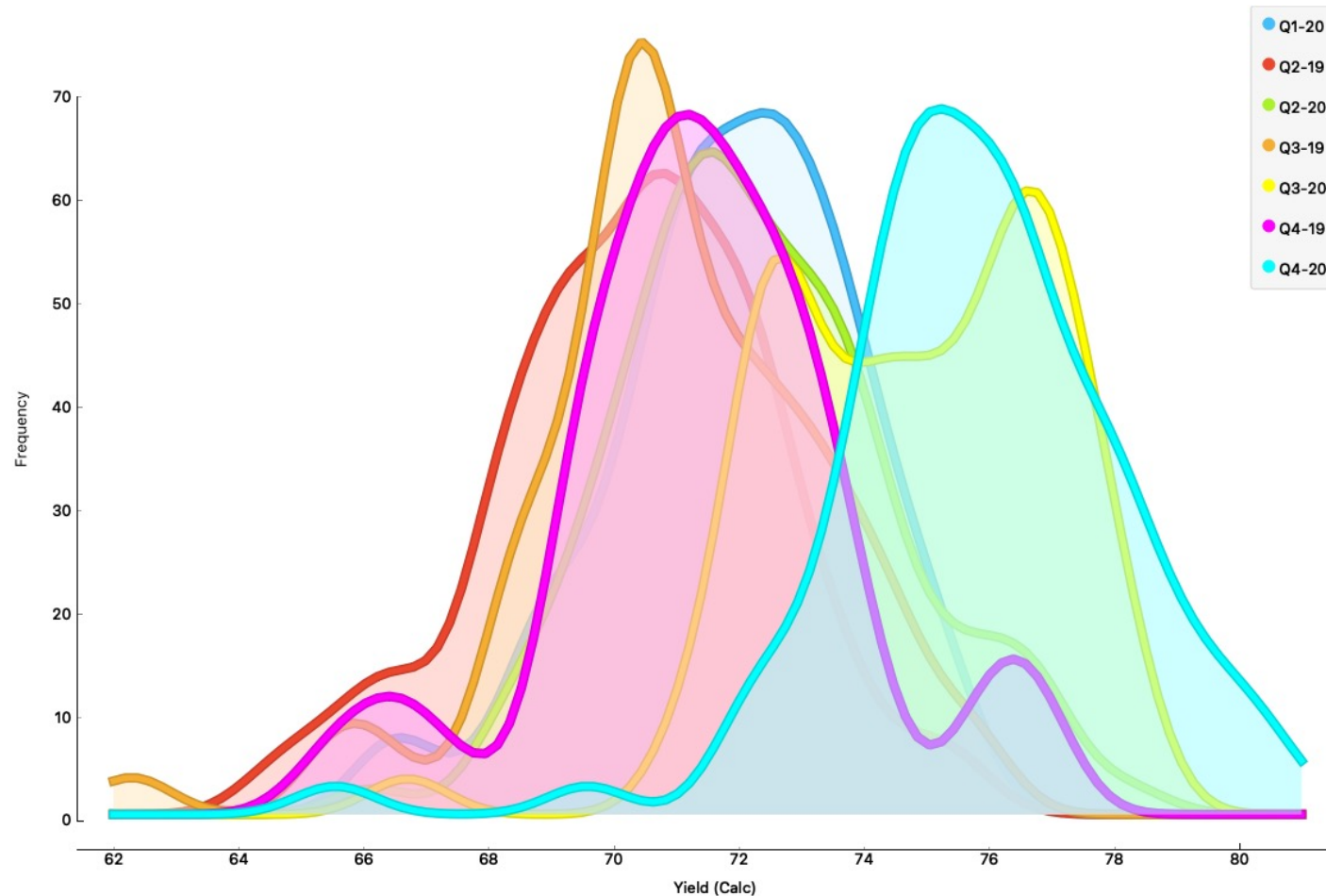
Challenges

- Uncountable Parameters likes Fines in natural Ra Material
- Interaction between the Parameters

Which Factors have highest Influence

Scoring Methods								
☒ Information Gain								
☒ Information Gain Ratio								
☒ Gini Decrease								
☒ ANOVA								
☒ χ^2								
☒ ReliefF								
☒ FCBF								
Select Attributes								
☐ None								
☐ All								
☐ Manual								
☒ Best ranked: 5								
☒ Send Automatically								
		#	Info. gain ∇	Gain ratio	Gini	ANOVA	χ^2	ReliefF
☒ Dosing Rate			0.242	0.121	0.063	73.436	105.275	0.082
☒ A4 LPH			0.208	0.126	0.054	58.937	75.627	0.052
☒ Temp-2			0.149	0.074	0.043	63.159	68.558	0.039
☒ Fill Temp			0.089	0.045	0.026	17.491	28.257	0.023
☒ Hold Time			0.087	0.043	0.033	5.273	8.131	0.051
☒ A1 %			0.069	0.035	0.019	41.373	36.189	0.040
☒ A2 %			0.052	0.026	0.014	28.043	28.934	0.028
☒ Fine TPH			0.039	0.020	0.011	11.097	20.628	0.017
☒ Fine			0.038	0.019	0.010	11.090	20.071	0.017
☒ A3%			0.036	0.022	0.010	1.162	0.848	0.031
☒ Coarse			0.031	0.016	0.011	3.851	3.327	0.015
☒ Coarse TPH			0.031	0.016	0.011	3.862	2.811	0.015

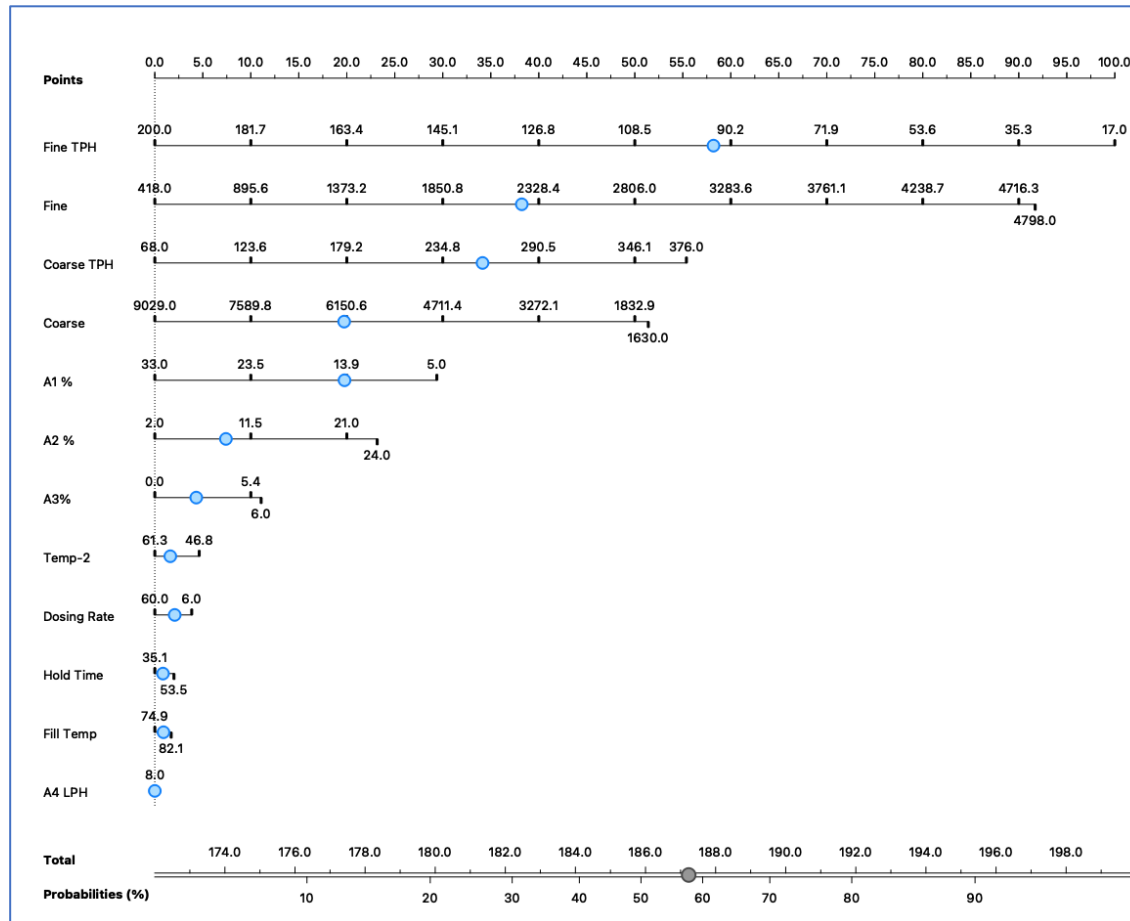
Variations in Yield due to Multiple Uncountable Parameters



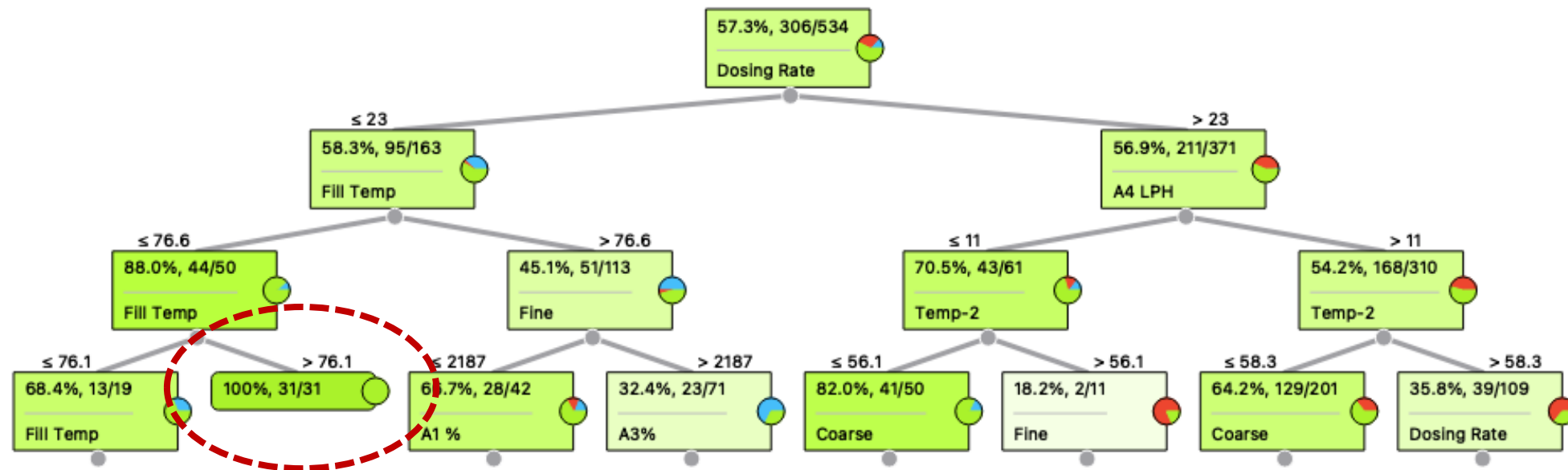
Relative Accuracy of Models

Sampling		Evaluation Results					
☒ Cross validation							
Number of folds:	20						
☒ Stratified							
☐ Cross validation by feature							
☐ Random sampling							
Repeat train/test:	10						
Training set size:	66 %						
☒ Stratified							
☐ Leave one out							
		Model	AUC	CA	F1	Precision	Recall
		Tree	0.880	0.869	0.869	0.869	0.869
		Random Forest	0.902	0.798	0.797	0.797	0.798
		Logistic Regression	0.750	0.639	0.627	0.632	0.639
		AdaBoost	0.861	0.850	0.850	0.850	0.850

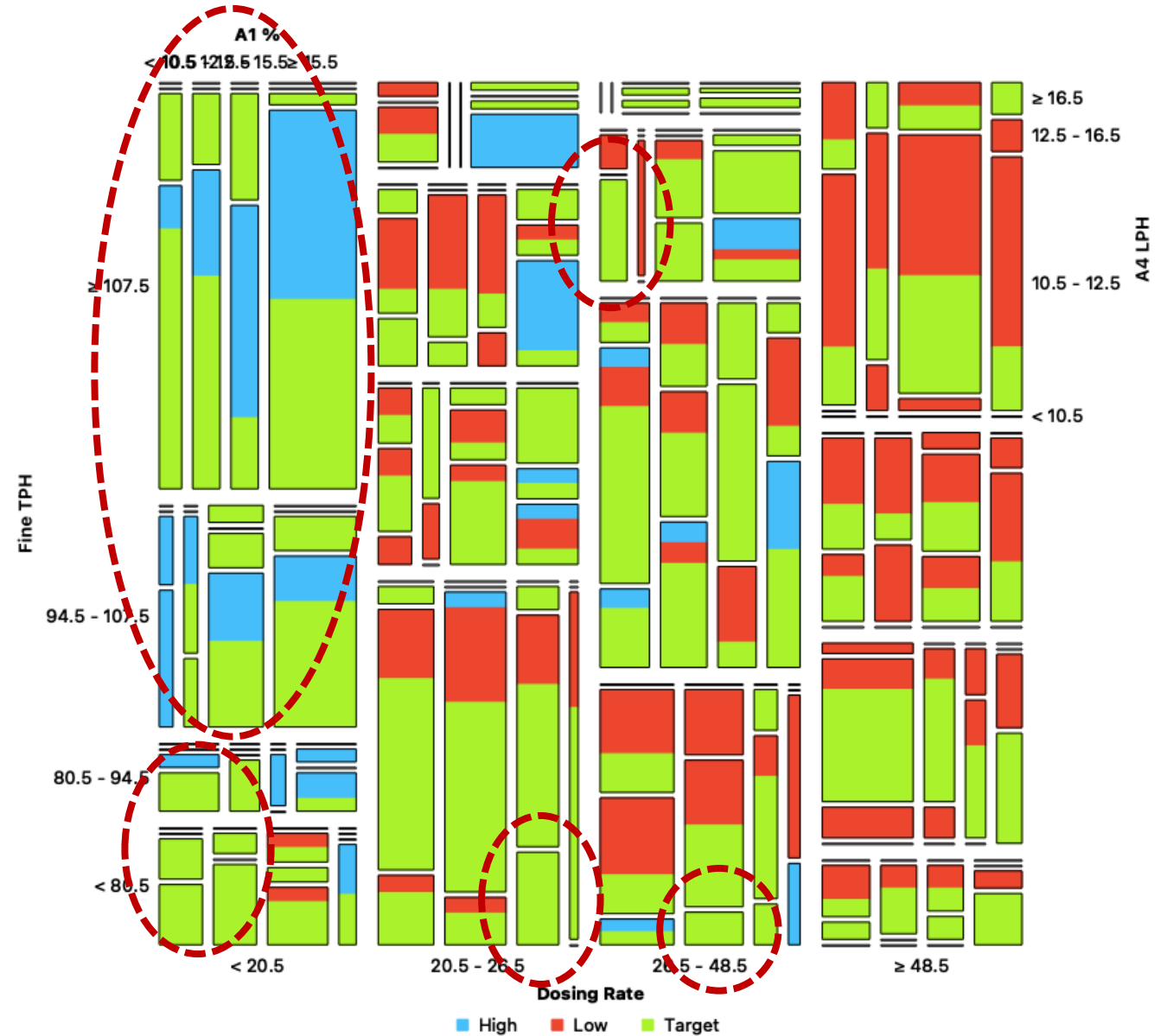
Predict Yield under Varying Operating Conditions



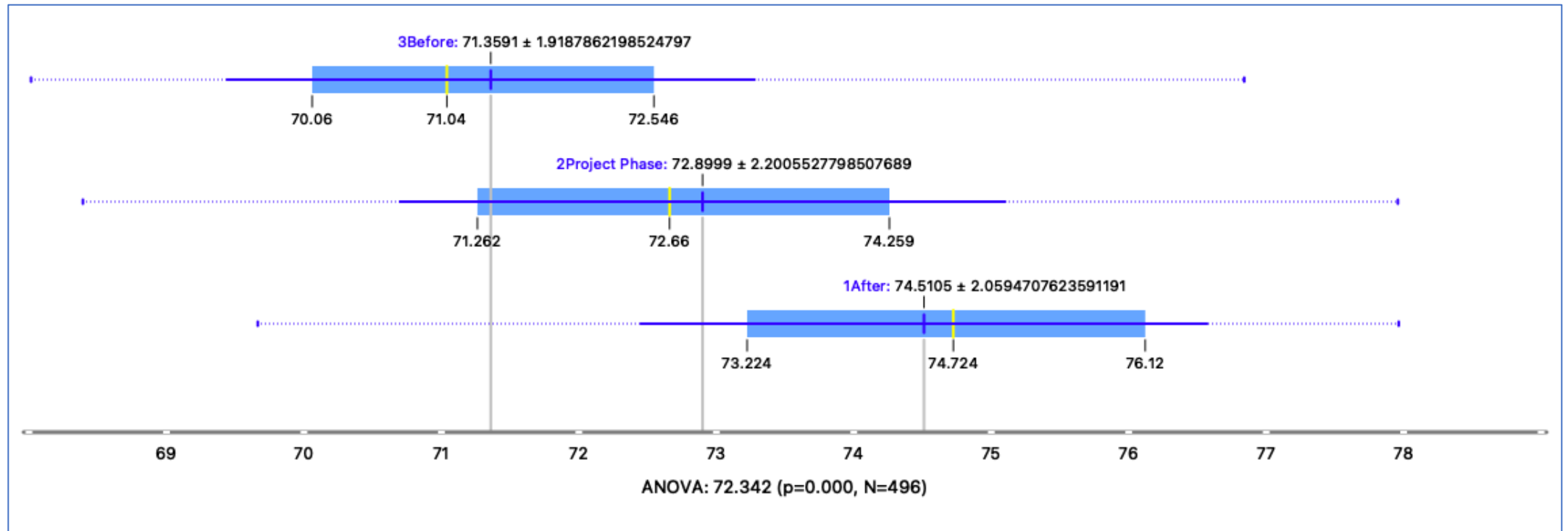
Factors driving Golden Batch(es)



Identify Optimal Solutions w/o Design of Experiments



Establish the Improvement with Visualisation





What Next
